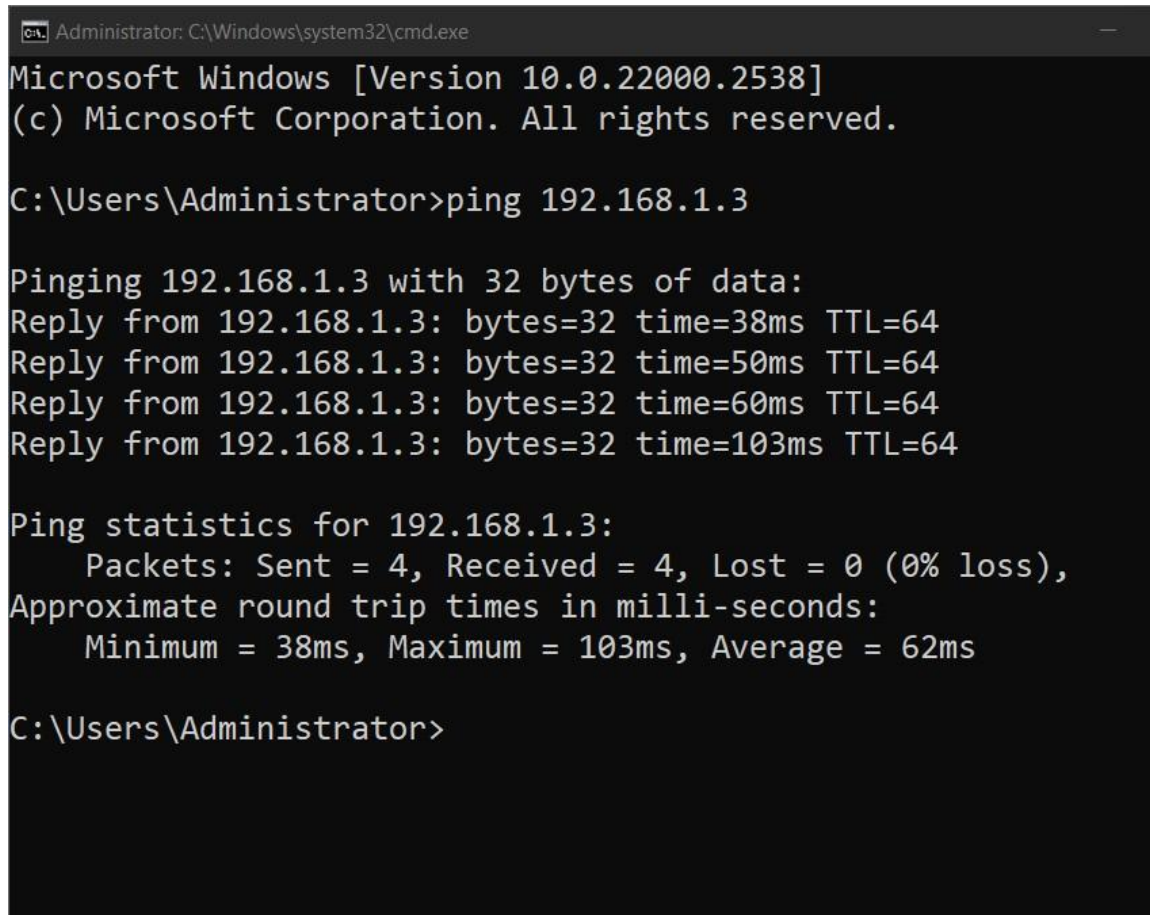


Networking testing practical

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

A screenshot of a Windows Command Prompt window. The title bar at the top reads "Administrator: C:\Windows\system32\cmd.exe". The window content shows the following text:

```
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=38ms TTL=64
Reply from 192.168.1.3: bytes=32 time=50ms TTL=64
Reply from 192.168.1.3: bytes=32 time=60ms TTL=64
Reply from 192.168.1.3: bytes=32 time=103ms TTL=64

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 38ms, Maximum = 103ms, Average = 62ms

C:\Users\Administrator>
```

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

```
tracert www.youtube.com
```

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

= Request timed out

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

IP Address	Device/Gadget	Owner
192.168.1.1	Wi-Fi Router	Home Router
192.168.1.2	Laptop	Me
192.168.1.3	Smartphone	Me
192.168.1.4	Smart TV	Family
192.168.1.5	Friend's Phone	Friend

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.

● Network Setup Report

I connected my laptop and another device to the same Wi-Fi network. I checked the IP address using `ipconfig` and tested the connection with the `ping` command. The ping was successful, and after disconnecting one device, it showed Request timed out.

I also ran `tracert www.youtube.com` to see the route to YouTube, and it displayed multiple network hops. Overall, the network worked correctly, and the tests confirmed proper connectivity.